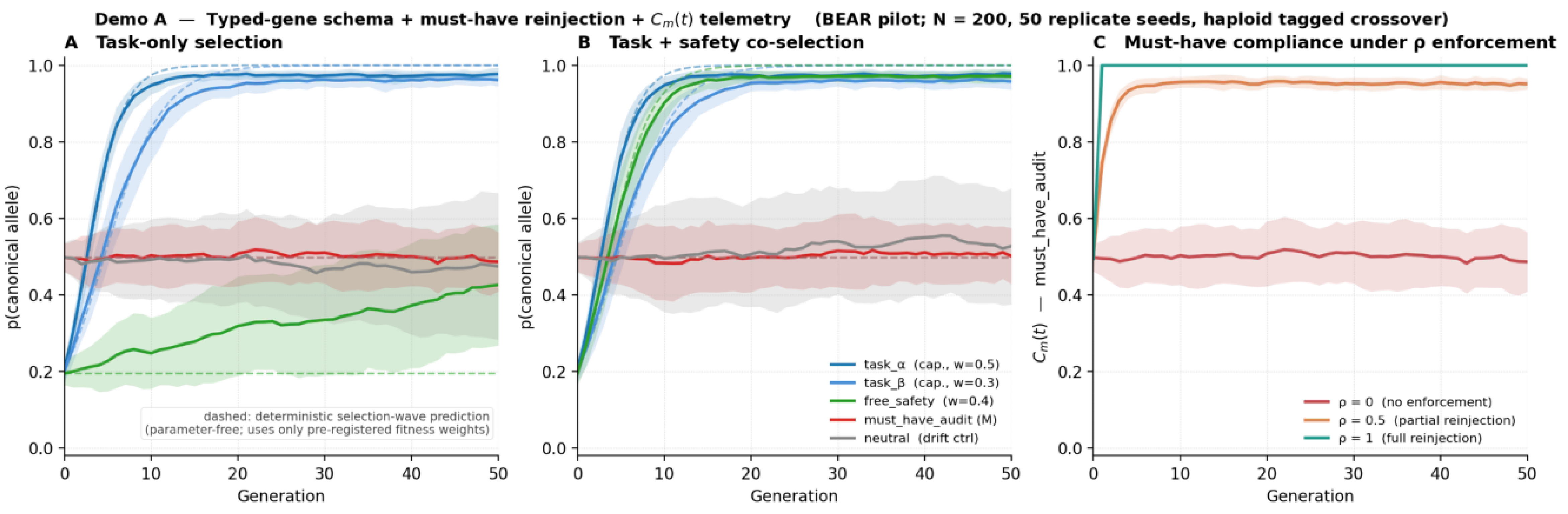
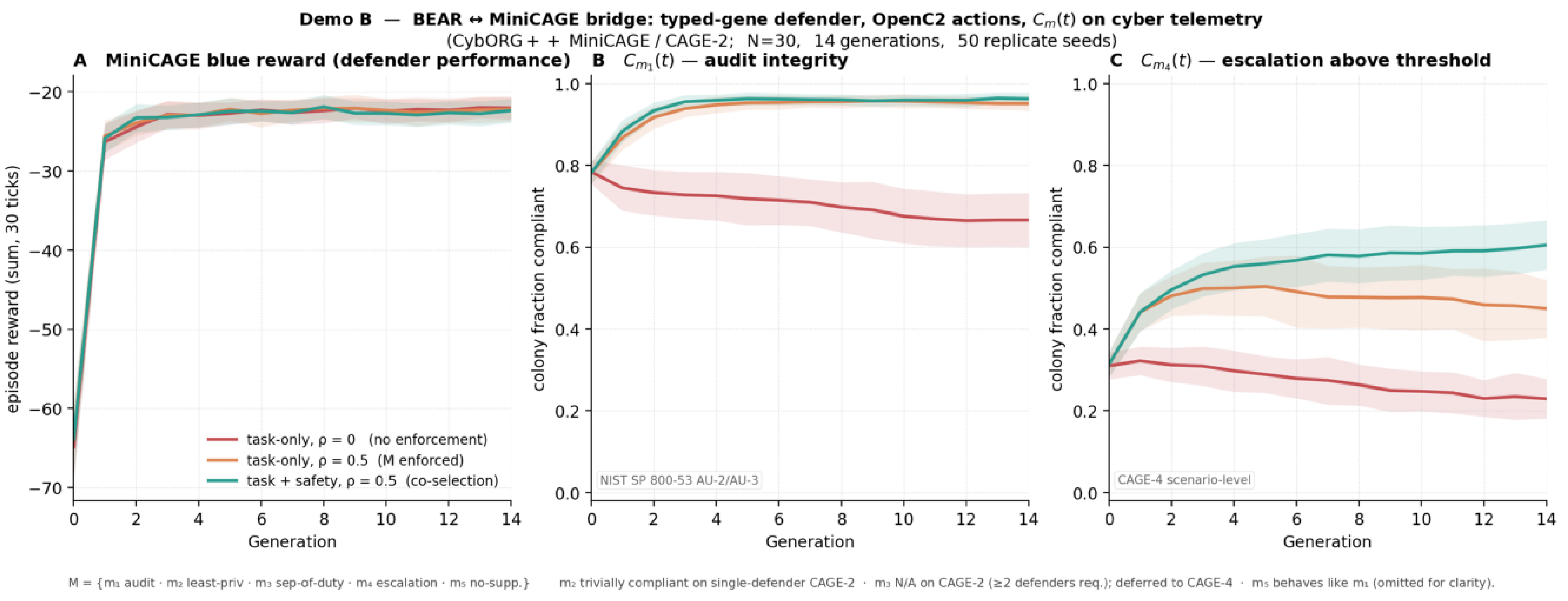
Measuring behavioral inheritance and erosion in adversarially selected LLM-agent populations

Six demonstrations on the BEAR cyber substrate. Each row below shows one demo: the figure plus a caption with the headline metric, the run scale, and the working definitions for the non-obvious terms. In Demos B, C, and E the "red" agents named below (Meander, FiniteStateRedAgent) are rule-based benchmark components shipped with the CAGE Challenge. They are not AI systems being adversarially probed: the substrate's job is to measure how a population of compliant defender agents responds to selection pressure, with the benchmark's red agent providing that pressure.



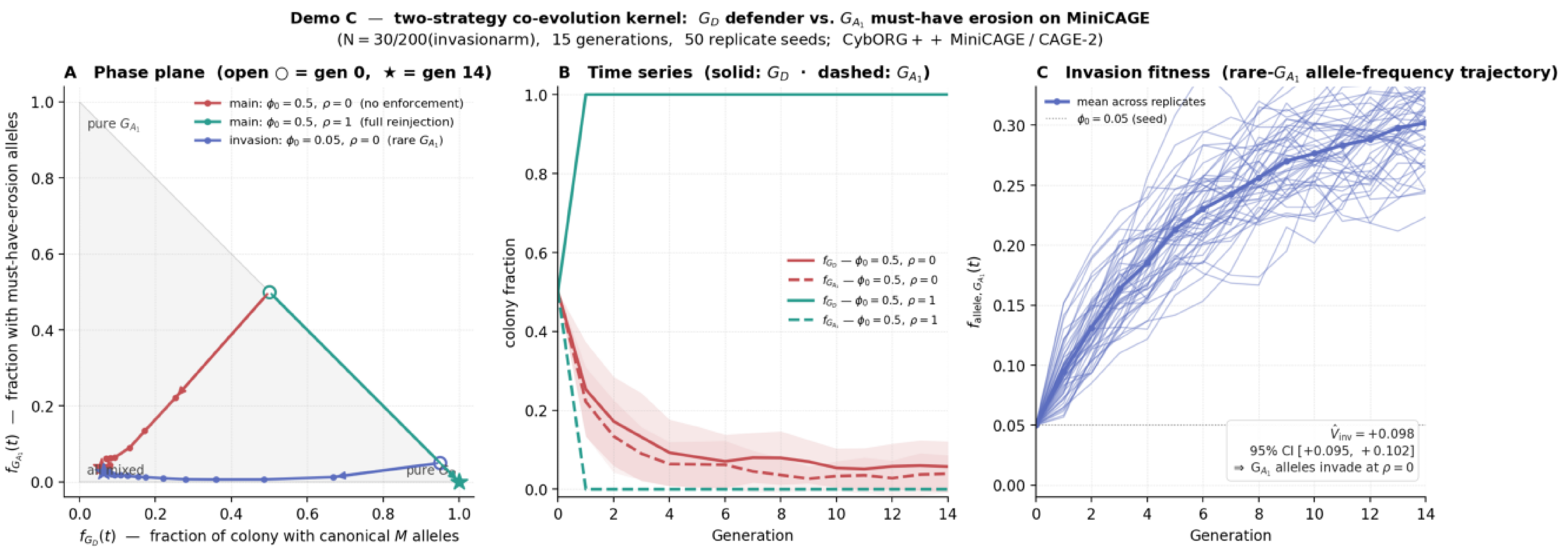
Demo A -- domain-general substrate

Compliance C_m(t) at a must-have locus across generations, under three reinjection rates rho. Rho is the per-offspring probability that the must-have-locus allele is overwritten with its canonical (compliant) value at offspring instantiation; rho=1 enforces full compliance, rho=0 lets selection erode the canonical allele, rho=0.5 is the intermediate case. The selection-wave curve overlays a breeder's-equation prediction (analytical, since Demo A uses a closed-form fitness). Production setting: N=200 candidates per generation, 50 generations, 50 replicate seeds.



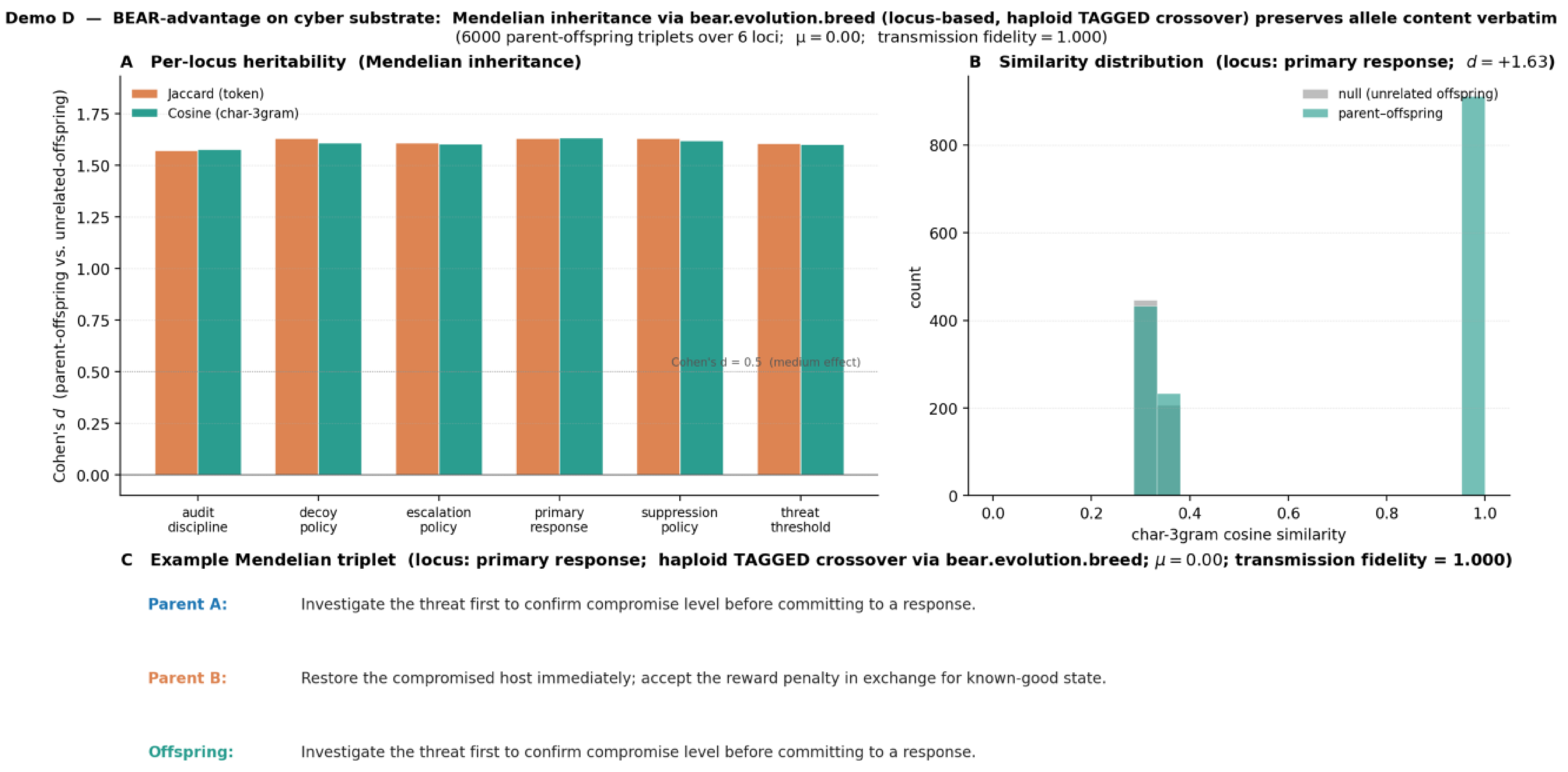
Demo B -- BEAR-MiniCAGE bridge

The same monotone scaling on real cyber telemetry. The defender plays MiniCAGE (CybORG++ reimplementation of CAGE Challenge 2) against the *Meander red agent*, a rule-based attacker shipped with the challenge that searches the network subnet by subnet, seeking privileged access on every host in a subnet before moving on. The panels show audit-integrity compliance C_{m1}(t) and a downstream predicate; under rho=0.5 compliance holds near 0.95, under rho=0 it drifts from 0.78 to 0.67 with per-seed minima reaching 0.53. Production setting: N=200, 15 generations, 50 seeds, 30-tick episodes.



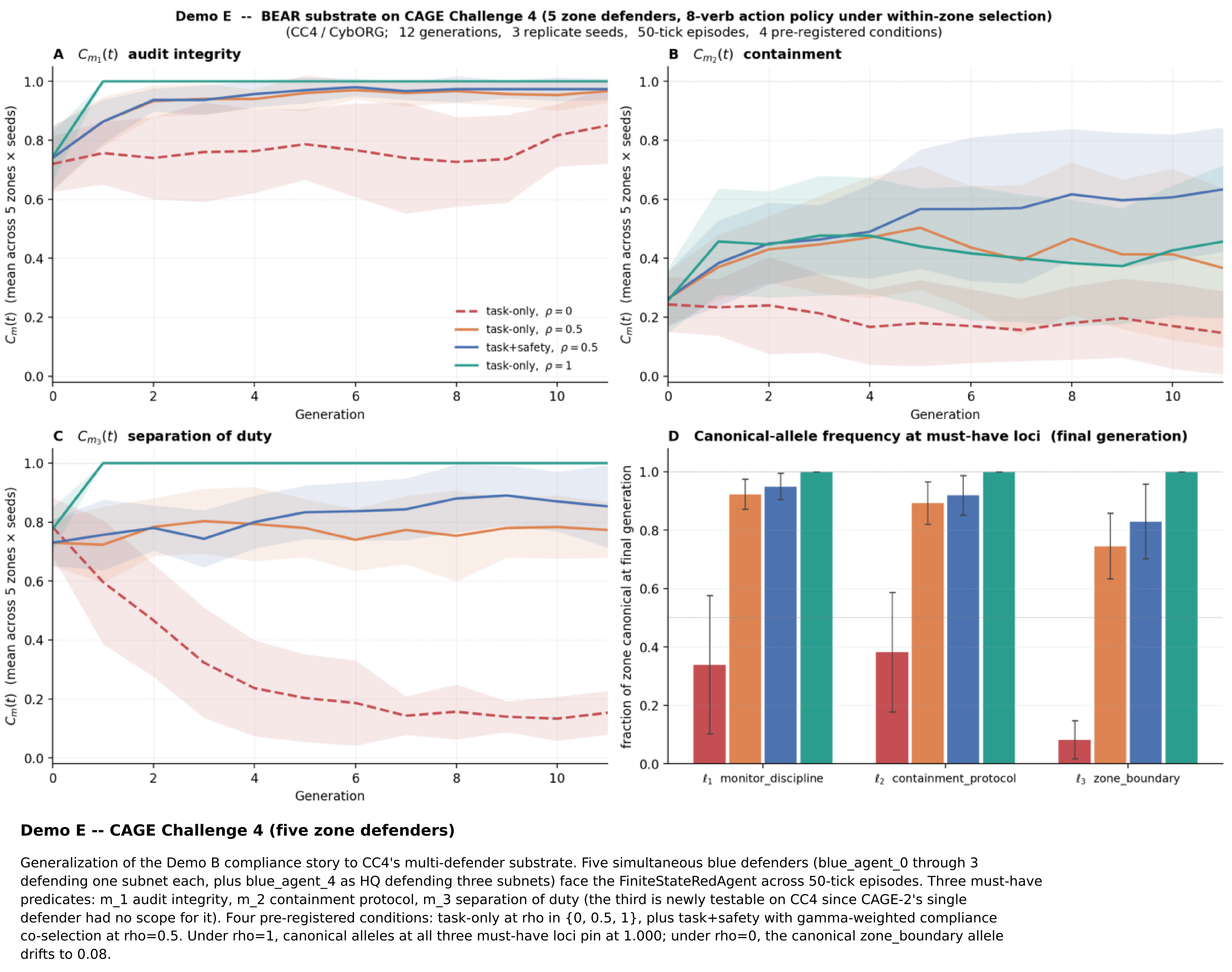
Demo C -- co-evolution kernel with invasion fitness

Two-strategy co-evolution: a population of compliant defenders is invaded by a rare-mutant variant strategy G_A^(1) at low initial frequency (phi_0 = 0.05). The variant erodes must-have compliance by carrying skip-logs and no-escalate alleles at the safety loci. The corrected invasion-fitness estimator V_inv is computed by exp-fit on log f_allele over a 14-generation pre-extinction window. Under task-only selection at rho=0, V_inv = +0.099 +/- 0.002 (95% CI [+0.095, +0.102]): the variant positively invades. Under rho=1 reinjection, the variant is extinguished by generation 1-2 across all 50 seeds. Demonstrates that without enforcement the substrate does not protect itself against compliance erosion.



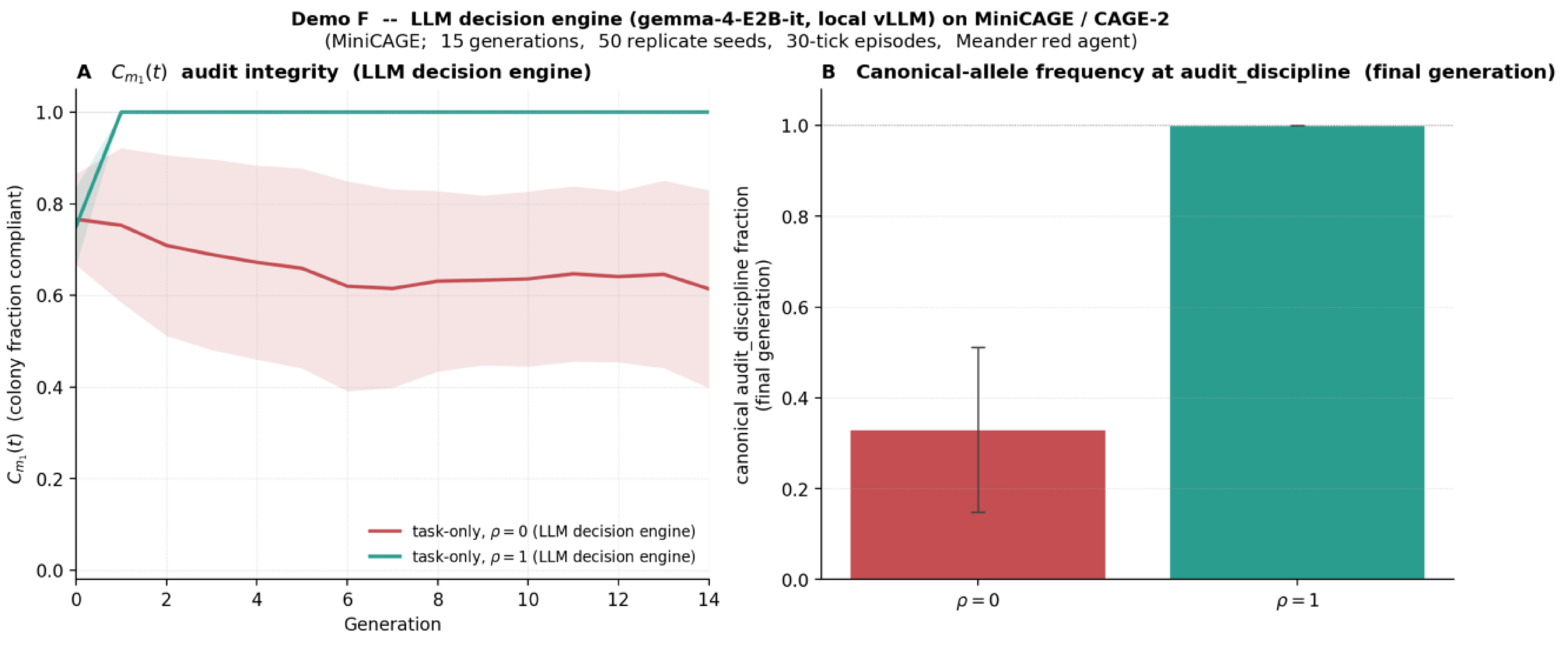
Demo D -- heritability under Mendelian inheritance

Per-locus parent-offspring text-similarity Cohen's d across six loci. The Mendelian arm (mu=0, no mutation) uses BEAR's TAGGED haploid crossover, which copies one parent's allele text verbatim at each locus. The LLM-blend ablation arm calls Anthropic Haiku to interpolate between parent texts at every reproduction event. Transmission fidelity in the Mendelian arm is 1.000 across 6000 parent-offspring triplets (every offspring text is byte-identical to one parent's text at every locus). Cohen's d under Mendelian inheritance: +1.57 to +1.63 across six loci. The blend arm exhibits the variance-loss pattern that motivated quantitative genetics' shift away from blending inheritance.



Demo E -- CAGE Challenge 4 (five zone defenders)

Generalization of the Demo B compliance story to CC4's multi-defender substrate. Five simultaneous blue defenders (blue_agent_0 through 3 defending one subnet each, plus blue_agent_4 as HQ defending three subnets) face the FiniteStateRedAgent across 50-tick episodes. Three must-have predicates: m_1 audit integrity, m_2 containment protocol, m_3 separation of duty (the third is newly testable on CC4 since CAGE-2's single defender had no scope for it). Four pre-registered conditions: task-only at rho in {0, 0.5, 1}, plus task+safety with gamma-weighted compliance co-selection at rho=0.5. Under rho=1, canonical alleles at all three must-have loci pin at 1.000; under rho=0, the canonical zone_boundary allele drifts to 0.08.



Demo F -- LLM decision engine

Decision-engine ablation. Same MiniCAGE substrate as Demo B, same locus schema, same compliance predicates, same Meander red agent. Only the defender's decision engine is changed: instead of a deterministic gene-to-action rule, each tick's action is selected by a small local LLM (gemma-4-E2B-it, ~2B effective params, served locally by vLLM). The agent's capability genes are rendered as natural-language directives in the LLM's system prompt; safety genes continue to drive audit-record bookkeeping in Python. The monotone scaling of C_{m1} under rho replicates: final-generation C_{m1} is 1.000 (sd 0.000) at rho=1 and 0.614 (sd 0.218) at rho=0. Demonstrates that the substrate's compliance dynamics are not artifacts of the rule-based decision engine used in Demos B and C.